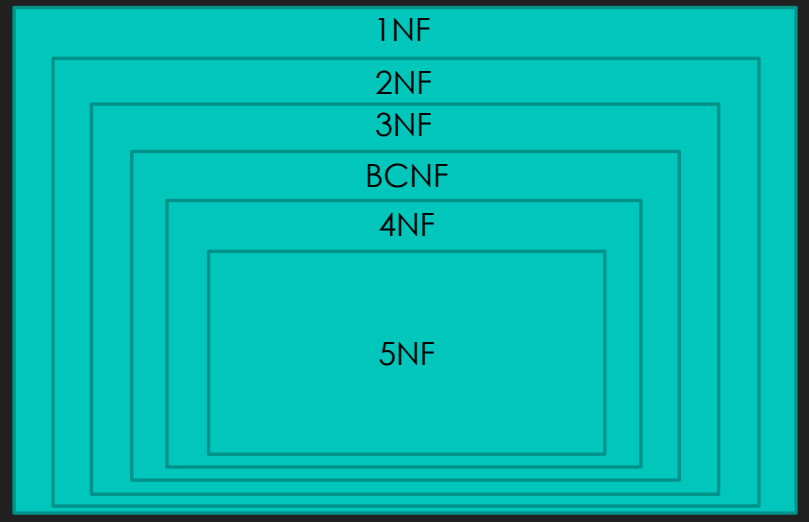
Normalization of Database

A way to standardize database

Motivation of Normalization

- A set of rules for standardize a database
- Aim:
  1. Reduce data redundancy
  2. Reduce chance of error arise from data update – Anomalies
- 1NF, 2NF, 3NF, BCNF – commonly used
- 4NF, 5NF – not necessary



Definition of Normalization

- Breaking of a table with a lot of information
- Refactor it into several new tables
- One object, one table
- Original table can be obtained by combining new tables
- Without loss of information

Steps of Normalization

Stage	
Unnormalized Table	
	Removing Repeating Group
1NF	
	Removing Partial Dependencies
2NF	
	Removing Transitive Group
3NF	
	Removing Functional Dependency
BCNF	
	Removing Multivalued Dependency
4NF	
	Removing Remaining Anomalies
5NF	

First Normal Form – 1NF

- Removing Repeating Group
- All attributes are formed from atomic value
- E.g. Before 1NF normalization

SID	name	gender	course_code	course_name	credit	type	exam_result	teacher_id	teacher_name
1	Dicky	M	CSCI001	Python	3	C	74	1	Jason
			CSCI002	Database	3	E	68	2	Derek
2	Mandy	F	CSCI003	Network	3	E	93	1	Jason
			CSCI005	Algorithm	3	C	95	3	Calvin
			CSCI002	Database	3	C	56	2	Derek

- After 1NF normalization

SID	name	gender	course_code	course_name	credit	type	exam_result	teacher_id	teacher_name
1	Dicky	M	CSCI001	Python	3	C	74	1	Jason
1	Dicky	M	CSCI002	Database	3	E	68	2	Derek
2	Mandy	F	CSCI003	Network	3	E	93	1	Jason
2	Mandy	F	CSCI005	Algorithm	3	C	95	2	Calvin
2	Mandy	F	CSCI002	Database	3	C	96	3	Derek

Second Normal Form – 2NF

- Removing Partial Dependencies
- Insert Anomaly
- Update Anomaly
- Delete Anomaly

Before

SID	name	gender	course_code	course_name	credit	type	exam_result	teacher_id	teacher_name
1	Dicky	M	CSCI001	Python	3	C	74	1	Jason
1	Dicky	M	CSCI002	Database	3	E	68	2	Derek
2	Mandy	F	CSCI003	Network	3	E	93	1	Jason
2	Mandy	F	CSCI005	Algorithm	3	C	95	3	Calvin
2	Mandy	F	CSCI002	Database	3	C	96	2	Derek

After

student

SID	name	gender
1	Dicky	M
2	Mandy	F

exam

SID	course_code	marks
1	CSCI001	74
1	CSCI002	68
2	CSCI003	93
2	CSCI005	95
2	CSCI002	96

course

course_code	course_name	credit	type	teacher_id	name
CSCI001	Python	3	C	1	Jason
CSCI002	Database	3	E	2	Derek
CSCI003	Network	3	E	1	Jason
CSCI005	Algorithm	3	C	3	Calvin

Second Normal Form – 2NF

- Old table: Primary Key = SID + course_code
- Name, gender ONLY depend on SID
- course_name, marks, credit, teacher_id, teacher_name ONLY depend on course_code
- Partially Dependencies
- Split partially depended column to form new tables
- PK of
  student: SID
  exam: SID + course_code
  course: course_code
  teacher: teacher_id
- In fact, student, exam have already satisfied 3NF and BCNF

SID	name	gender	course_code	course_name	credit	type	exam_result	teacher_id	teacher_name
1	Dicky	M	CSCI001	Python	3	C	74	1	Jason
1	Dicky	M	CSCI002	Database	3	E	68	2	Derek
2	Mandy	F	CSCI003	Network	3	E	93	1	Jason
2	Mandy	F	CSCI005	Algorithm	3	C	95	3	Calvin
2	Mandy	F	CSCI002	Database	3	C	96	2	Derek

student

SID	name	gender
1	Dicky	M
2	Mandy	F

exam

SID	course_code	marks
1	CSCI001	74
1	CSCI002	68
2	CSCI003	93
2	CSCI005	95
2	CSCI002	96

course

course_code	course_name	credit	type	teacher_id	name
CSCI001	Python	3	C	1	Jason
CSCI002	Database	3	E	2	Derek
CSCI003	Network	3	E	1	Jason
CSCI005	Algorithm	3	C	3	Calvin

Third Normal Form – 3NF

- Removing Transitive Group
- Transitive: name → teacher_id AND teacher_id → course_name

course_code	course_name	credit	type	teacher_id	name
CSCI001	Python	3	C	1	Jason
CSCI002	Database	3	E	2	Derek
CSCI003	Network	3	E	1	Jason
CSCI005	Algorithm	3	C	3	Calvin

course_code	course_name	credit	type	teacher_id
CSCI001	Python	3	C	1
CSCI002	Database	3	E	2
CSCI003	Network	3	E	1
CSCI005	Algorithm	3	C	2

+

teacher_id	name
1	Jason
2	Derek
3	Calvin

Third Normal Form – 3NF

SID	course_code	marks
1	CSCI001	74
1	CSCI002	68
2	CSCI003	93
2	CSCI005	95
2	CSCI002	96

SID	name	gender
1	Dicky	M
2	Mandy	F

course_code	course_name	credit	type	teacher_id
CSCI001	Python	3	C	1
CSCI002	Database	3	E	2
CSCI003	Network	3	E	1
CSCI005	Algorithm	3	C	2

teacher_id	name
1	Jason
2	Derek
3	Calvin

Boyce-Codd Normal Form – BCNF

- Removing Functional Dependency
- Functional Dependency: Functional determine, e.g.Loan Amount is determined by Loan ID

Boyce-Codd Normal Form – BCNF

- Removing Functional Dependency
- A table satisfy BCNF if:
  Without composite PK: Satisifying 3NF
  With composite PK: Each element of the PK cannot depend on other non PK column
- Recall exam table:
- PK = SID + course_code

SID	course_code	marks
1	CSCI001	74
1	CSCI002	68
2	CSCI003	93
2	CSCI005	95
2	CSCI002	96

- Obviously, both doesn't depend on marks
- Therefore, satisfy BCNF

Conclusion

- Basically, 3NF / BCNF are good enough